



IX Studio CR: Code Test Description

Back End Position - Node.js

At ImagineX Studio - Costa Rica, we truly believe in the importance of finding the correct talent, not only tech-savvy but also a great team player.

We enjoy writing great software, and we love creating solutions to different problems with our peers.

Evaluation description

In order to evaluate some aspects from our candidates, we design a code test where the interviewees will be able to apply their expertise and share with us some of their skills. We value clean and well code-structured code, creativity and problem solving.

Evaluation topic

The interviewee will develop an application using Node.js/TypeScript. Feel free to propose any framework/library you find optimal for the solution.

It's required to build a "Clock" application that will print messages on screen at specific time intervals.

Acceptance criteria

- Print the following values at the intervals indicated:
 - "tick" - every second.
 - "tock" - every minute.
 - "bong" - every hour.
- Only one value should be printed in a given second.

- i.e. when printing "bong" on the hour, the "tick" and "tock" values should not be printed.
- The app should run for three hours and then exit.
- A mechanism should exist for the user **to alter any of the printed values while the program is running.**
 - i.e. after the clock has run for 10 minutes I should, without stopping the program, be able to change it so that it stops printing "tick" every second and starts printing "quack" instead.
- **Provide appropriate test coverage.**
- We encourage you to structure the project using the best practices that you know, preparing your project for a production-ready version that will be supported by another team.

Your completed code test should be published within a private GitHub repo where the user **ix-code-reviewer** (<https://github.com/ix-code-reviewer>) is set as a collaborator.

Technical considerations

We encourage interviewees to implement and take into consideration the following points:

- Implement your project using Node.js/TypeScript.
 - Your repository should be self-explanatory, create a great **README.md** document.
 - We want to see best practices, structure your project as you would have been creating a project to be deployed in production.
 - Review that your project has all the required dependencies to be installed and works without any issue.
 - Testing is always important, try to implement at least basic testing.

Evaluation criteria

1. Interviewees will be presented with a problem. They must understand the problem and come up with a solution.
2. Problem solving skills. We expect the candidates can expose their ideas clearly to the team and clients, hence communication is a key skill to have.
3. As software engineers, we expect the interviewees to use all the resources at hand. We focus on delivering solutions to our clients, therefore we try to take the most advantage of our experience developing software.

4. Assure that logic runs as expected is very important when we develop software. Testing techniques will be a topic to be discussed during the session.